

JAVA PROGRAMMING LAB RECORD

Basic Information	
Experiment No	02
Date	21/09/26
Title	Implementation of NavigableSet Methods using TreeSet in Java

Objective / Aim

To write and execute a Java program that demonstrates the commonly used methods of the NavigableSet interface using TreeSet, including lower(), floor(), ceiling(), higher(), pollFirst(), pollLast(), and descendingSet().

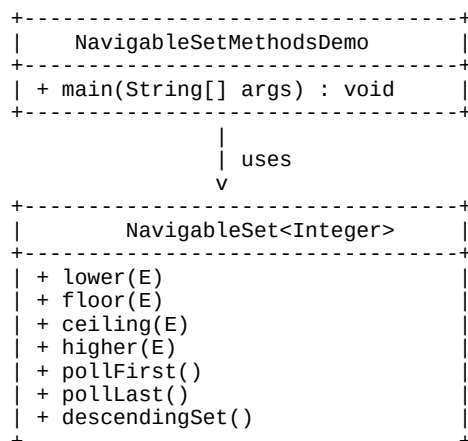
System Requirements

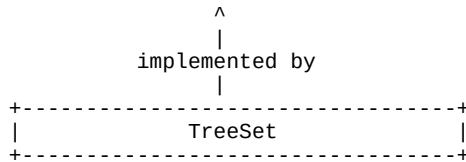
Hardware Requirements	Software Requirements
Computer/Laptop	Operating System: Windows / Linux / macOS
Processor: Intel/AMD dual-core or above	Java Development Kit (JDK): 17 or later
RAM: 4 GB or above	IDE: Eclipse / IntelliJ IDEA / NetBeans / VS Code
Storage: At least 1 GB free space	Command Prompt / Terminal

Algorithm / Logic Description

1. Import NavigableSet from java.util and TreeSet from java.util.
2. Create a NavigableSet of Integer values using TreeSet.
3. Add the values 10, 20, 30, 40, 50, and 60 to the set.
4. Display the elements of the NavigableSet in ascending order.
5. Use lower(30) to find the greatest element strictly less than 30.
6. Use floor(30) to find the greatest element less than or equal to 30.
7. Use ceiling(35) to find the smallest element greater than or equal to 35.
8. Use higher(30) to find the smallest element strictly greater than 30.
9. Use pollFirst() to remove and return the first element.
10. Use pollLast() to remove and return the last element.
11. Use descendingSet() to display the remaining elements in descending order.
12. Stop the program.

UML Class Diagram





Source Code

```

import java.util.NavigableSet;
import java.util.TreeSet;

class NavigableSetMethodsDemo {

    public static void main(String[] args) {

        // Creating a NavigableSet using TreeSet
        NavigableSet<Integer> values = new TreeSet<>();

        // Adding elements
        values.add(10);
        values.add(20);
        values.add(30);
        values.add(40);
        values.add(50);
        values.add(60);

        System.out.println("NavigableSet: " + values);

        // lower()
        System.out.println("Lower than 30: " + values.lower(30));

        // floor()
        System.out.println("Floor of 30: " + values.floor(30));

        // ceiling()
        System.out.println("Ceiling of 35: " + values.ceiling(35));

        // higher()
        System.out.println("Higher than 30: " + values.higher(30));

        // pollFirst()
        System.out.println("Removed first element: " + values.pollFirst());

        System.out.println("After pollFirst(): " + values);

        // pollLast()
        System.out.println("Removed last element: " + values.pollLast());

        System.out.println("After pollLast(): " + values);

        // descendingSet()
        System.out.println("Descending Set: " + values.descendingSet());
    }
}

```

Completion and Execution Commands

1. Save the program with the file name:
NavigableSetMethodsDemo.java
2. Open Command Prompt / Terminal and move to the folder containing the file:
cd path_to_your_folder
3. Compile the Java program:
javac NavigableSetMethodsDemo.java

4. Execute the compiled program:
java NavigableSetMethodsDemo

Output

```
NavigableSet: [10, 20, 30, 40, 50, 60]
Lower than 30: 20
Floor of 30: 30
Ceiling of 35: 40
Higher than 30: 40
Removed first element: 10
After pollFirst(): [20, 30, 40, 50, 60]
Removed last element: 60
After pollLast(): [20, 30, 40, 50]
Descending Set: [50, 40, 30, 20]
```

Result / Completion

The Java program was successfully implemented and executed. The NavigableSet methods lower(), floor(), ceiling(), higher(), pollFirst(), pollLast(), and descendingSet() were demonstrated using a TreeSet.